

Project Design Document: HiClickMe

1. Project Overview

Project Name: HiClickMe (Multi-Tenant URL Shortener & Analytics SaaS)

Purpose: Provide a scalable, multi-tenant URL shortening platform with enterprise subdomains, QR codes, UTM tracking, and premium subscription options.

Goals:

- Instant URL redirection (<10ms for cached links)
 - Asynchronous analytics logging for clicks, geolocation, and device data
 - Multi-tenant support with custom subdomains and branding
 - Premium plans with payments and feature gating
 - Cloud-deployed, scalable architecture using serverless and managed services
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2. System Architecture

2.1 High-Level Components

Component	Purpose	Technology
Frontend Dashboard	User management, link creation, analytics	Next.js, React, shadcn UI
Redirect Server	Handles URL redirection & optional preview	Node.js (Express) on AWS Lambda / EC2
Backend API	CRUD for URLs, analytics logging, premium features	Node.js (Express/Fastify) or FastAPI
Database 1	User & subscription data	PostgreSQL (RDS / Aurora)
Database 2	Key-value URL mapping & analytics	DynamoDB
Cache	Hot link lookups, sessions, rate-limiting	Redis (Elasticache / managed)
Storage	QR code images, preview page assets	S3 + CloudFront CDN
Payment Gateway	Premium subscription payments	Stripe / PayPal
CI/CD Pipeline	Automated testing & deployment	GitHub Actions / AWS CodePipeline / Terraform

2.2 Architectural Diagram



3. Data Model

3.1 PostgreSQL (User & Subscription Data)

```
sql
```

```
CREATE TABLE users (  
  id SERIAL PRIMARY KEY,  
  email VARCHAR(255) UNIQUE NOT NULL,  
  password_hash TEXT NOT NULL,  
  plan VARCHAR(50) DEFAULT 'free',  
  created_at TIMESTAMP DEFAULT NOW()  
);  
  
CREATE TABLE subscriptions (  
  id SERIAL PRIMARY KEY,  
  user_id INT REFERENCES users(id),  
  stripe_subscription_id TEXT,  
  status VARCHAR(50),  
  start_date TIMESTAMP,  
  end_date TIMESTAMP  
);
```

3.2 DynamoDB (URL Mapping & Analytics)

- **Table:** URLMappings
 - Partition Key: `shortCode`
 - Sort Key: `tenantId`
 - Attributes: `destinationUrl`, `createdAt`, `clickCount`, `metadata`
- **Table:** ClickAnalytics
 - Partition Key: `shortCode`
 - Sort Key: `timestamp`
 - Attributes: `userAgent`, `geo`, `referrer`, `utm`

4. Key Features

4.1 URL Shortening

- Generate unique short codes
- Custom codes for premium users
- Store mapping in DynamoDB

4.2 Redirect

- Instant redirect (<10ms with Redis cache)
- Optional preview/confirmation page
- Fire-and-forget analytics logging

4.3 Analytics

- Track clicks, geolocation, device type, referrer, UTM params
- Async updates to DynamoDB / Postgres
- Dashboard visualization

4.4 Multi-Tenant / Enterprise

- Custom subdomains: `client1.hiclickme.com`
- Role-based access per tenant
- Optional white-labeled QR codes

4.5 QR Code Generation

- Generate QR code for each short URL
- Store PNG/SVG on S3
- Serve via CDN

4.6 Premium Features / Payments

- Stripe integration for subscriptions
 - Feature gating (custom subdomain, branded QR codes, higher API limits)
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5. Performance & Caching Strategy

- **Redis LRU cache** for hot links
 - Cache TTL: 5–15 mins
 - Cold links → DynamoDB → populate cache
 - Async analytics to avoid slowing redirects
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6. Deployment Strategy

- **Frontend:** Vercel or S3 + CloudFront
 - **Backend / API:** AWS Lambda (serverless) or EC2
 - **Databases:** Postgres (RDS/Aurora) for users, DynamoDB for links
 - **Caching:** Redis Elasticache
 - **Storage:** S3 for QR codes / preview assets
 - **CI/CD:** GitHub Actions → automated deploy to Lambda/EC2 + frontend
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7. Security & Reliability

- JWT-based authentication
 - OAuth2 / Google login for users
 - HTTPS enforced via CloudFront / ALB
 - Rate-limiting per tenant & IP
 - Daily backups for Postgres & DynamoDB
 - Optional monitoring via CloudWatch / Sentry
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8. Optional Advanced Features

- Bulk link import / export for enterprise tenants
 - UTM parameter generation & analytics
 - Admin dashboard for usage metrics across tenants
 - Cron jobs to pre-warm hot links in Redis
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✅ **HiClickMe is now positioned as a full-featured, production-ready, multi-tenant SaaS URL shortener with cloud deployment, caching, analytics, and premium subscription features.**